CSE2023-HW5

(1) $a_n - 3a_{n-1} - a_{n-2} + 3a_{n-3} = 0$

CE: $x^3 - 3x^2 - x + 3 = 0 \rightarrow (x-3)(x+1)(x-1) = 0 \quad x = 3, -1, 1$

$a_n = A(x_1)^n + B(x_2)^n + C(x_3)^n \rightarrow a_n = A(3)^n + B(-1)^n + C(1)^n$

$a_0 = 4 = A + B + C$

$a_1 = 10 = 3A - B + C$

$a_2 = 12 = 9A + B + C$

$4A + 2C = 14$

$12A + 2C = 22$

$8A = 8 \rightarrow A = 1$

$C = 5$

$B = -2$

$$a_n = 3^n - 2(-1)^n + 5(1)^n$$

(2) $a_n = 3a_{n-1} + 5 \cdot 3^n$

CE: $r = 3$

$a_n^{(h)} = \alpha \cdot 3^n$

$F(n) = 5\,3^n = 5 \cdot n^0 \cdot 3^n$ the root is 3

$a_n^{(p)} = n \cdot c \cdot 3^n$

$n \cdot c \cdot 3^n = 3 \cdot (n-1) \cdot c \cdot 3^{n-1} + 5 \cdot 3^n$

$nc \cdot 3^n = (n-1)c \cdot 3^n + 5 \cdot 3^n$

$nc \cdot 3^n = nc \cdot 3^n - c3^n + 5 \cdot 3^n$

$c \cdot 3^n = 5 \cdot 3^n$

$c = 5$

$a_n^{(p)} = 5n \cdot 3^n$

$a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = \alpha \cdot 3^n + 5n \cdot 3^n$$

(3) a) $a_n = 2a_{n-1} + 150(n-1)$

b) CE: $r-2=0 \rightarrow r=2$

$a_n^{(h)} = \alpha \cdot 2^{n-1}$ $\quad a_n^{(p)} = Cn + d$

$Cn+d = 2\{C(n-1)+d\} + 150n - 150 = 2Cn - 2C + 2d + 150n - 150$

$C = -150$ $\quad d = 2d - 2C - 150$

$d = 2C + 150$

$d = -150$

$a_n^{(p)} = -150n - 150$

$a_n = a_n^{(h)} + a_n^{(p)} = \alpha \cdot 2^{n-1} - 150n - 150$

$a_1 = 2000 = \alpha \cdot 2^0 - 150 - 150$

$\alpha = 2300$

$$a_n = 2300 \cdot 2^{n-1} - 150n - 150$$

(4) A: set of all people with doctorates

$(a,b) \in R^2 \leftrightarrow$ There is a person x_1 in A such that 'a' is the advisor of x_1 and x_1 is the advisor of b.

for any $n \in Z^+$

$(a,b) \in R^n \leftrightarrow$ There is sequence of persons $x_1, x_2, \cdots, x_{n-1}$ in A such that 'a' is the advisor of x_1, x_{n-1} is the advisor of b and x_i is the advisor of x_{i+1} for all $1 \leq i \leq n-2$

(5) If R is a relation defined on a finite set A, then

$$R^{-1} = \{(b,a) / (a,b) \in R\}$$

Suppose R is a relation defined on a set $A = \{a_1, a_2, \cdots, a_n\}$

Then R can be represented by the matrix $M_R = [m_{ij}]$

$$m_{ij} = \begin{cases} 1 & , (a_i, a_j) \in R \\ 0 & , (a_i, a_j) \notin R \end{cases}$$

Because $(a_i, a_j) \in R$ if and only if $(a_j, a_i) \in R^{-1}$, and $m_{ij} = 1$ in M_R if and only if $m_{ij} = 1$ in $M_{R^{-1}}$

Thus, $M_{R^{-1}}$ can be found from M_R by writing 1 in $(i,j)^{th}$ position whenever $(j,i)^{th}$ position is 1 in M_R and writing 0 in all other positions.